ANACONDA & PYTHON

PROGRAMMING

Complete Question Paper with Answer Key

Comprehensive Study Material

SECTION A: Objective Type Questions (MCQs)

Q1. Who developed and maintains the Anaconda distribution?

A. Guido van Rossum
B. John D. Hunter
C. Peter Wang and Travis Oliphant
D. Travis Oliphant and Guido van Rossum
✓ *Answer: C*

Q2. Which command is used in the Anaconda prompt to activate a newly created virtual environment named myenv?

A. conda start myenv
B. conda activate myenv
C. activate conda myenv
D. conda run myenv
✓ *Answer: B*

Q3. What is the default URL port when Jupyter Notebook opens in a web browser?

A. localhost:8888
B. localhost:8080
C. localhost:8000
D. localhost:9999
✓ *Answer: A*

Q4. In Jupyter Notebook, which menu allows you to restart or shut down the computational engine?

A. File Menu
B. Run Menu
C. Settings Menu
D. Kernel Menu
✓ *Answer: D*

Q5. Python was designed at Centrum Wiskunde & Informatica in which year?

A. 2012
B. 1991
C. 1989
D. 2003
✓ *Answer: B*

Q6. Which of the following is NOT a valid identifier in Python?

A. _myVar
B. my_var2
C. 2myVar
D. myVar_
✓ *Answer: C*

Q7. Which of the following is a mutable Sequence Data Type in Python?

A. Tuple
B. String
C. List
D. Range

✓ *Answer: C*

Q8. What is the result of the expression 10 // 3 in Python?

A. 3.33
B. 3
C. 1
D. 4
✓ *Answer: B*

Q9. Which operator is used for Exponentiation in Python?

A. ^
B. //
C. **
D. %
✓ *Answer: C*

Q10. What is the output of 'Hello' * 3?

A. Hello Hello Hello
B. Hello3
C. HelloHelloHello
D. Error
✓ *Answer: C*

Q11. According to Operator Precedence, which of the following is evaluated first?

A. Addition +
B. Multiplication *
C. Parentheses ()
D. Exponentiation **
✓ *Answer: C*

Q12. What is the ASCII code for the uppercase letter 'A'?

A. 97
B. 32
C. 65
D. 48
✓ *Answer: C*

Q13. Which function converts a string to an integer explicitly?

A. float()
B. str()
C. int()
D. bool()
✓ *Answer: C*

Q14. By default, what does the print() function append at the end of the output?

A. A space
B. A newline
C. A tab
D. A comma
✓ *Answer: B*

Q15. What data type does the input() function return by default?

A. Integer
B. Float
C. String
D. Boolean
✓ *Answer: C*

Q16. Which keyword is used to check multiple conditions if the first if condition is False?

A. else if
B. elseif
C. elif
D. else
✓ *Answer: C*

Q17. A directory containing multiple Python modules is known as a:

A. Package
B. Virtual Environment
C. Function
D. Dictionary
✓ *Answer: A*

Q18. Which library provides support for multi-dimensional arrays and matrices?

A. Pandas
B. NumPy
C. SciPy
D. Matplotlib
✓ *Answer: B*

Q19. Which of the following commands creates a 1-D NumPy array?

A. np.array([1, 2, 3])
B. np.ones((2,3))
C. np.array([[1,2], [3,4]])
D. np.zeros((2,3))
✓ *Answer: A*

Q20. Compared to Python Lists, NumPy arrays are:

A. Heterogeneous
B. Slower
C. More memory efficient
D. Able to store mixed data types natively
✓ *Answer: C*

Q21. If a = np.array([10, 20]) and b = np.array([2, 5]), what is a / b?

A. [12, 25]
B. [5.0, 4.0]
C. [8, 15]
D. [20, 100]
✓ *Answer: B*

Q22. Which NumPy function returns the shape (dimensions) of an array?

A. np.size

B. np.ndim
C. np.shape
D. np.dtype
✓ *Answer: C*

Q23. Which function returns the index of the maximum value in a NumPy array?

A. np.max
B. np.mean
C. np.argmax
D. np.argmin
✓ *Answer: C*

Q24. The name 'Pandas' is derived from:

A. Panel Data
B. Python Data Analysis
C. Parallel Arrays
D. Python And Data Structures
✓ *Answer: A*

Q25. A Pandas Series is best described as:

A. A multi-dimensional unlabelled array
B. A one-dimensional labeled array
C. A two-dimensional table
D. A graphical plot
✓ *Answer: B*

Q26. In Pandas, which attribute returns the number of elements in a Series?

A. .size
B. .values
C. .index
D. .empty
✓ *Answer: A*

Q27. Which function returns the first n rows of a Pandas Series?

A. .tail(n)
B. .first(n)
C. .head(n)
D. .top(n)
✓ *Answer: C*

Q28. How do you delete a column named 'City' from a Pandas DataFrame?

A. df.remove('City')
B. df.drop(columns='City', inplace=True)
C. df.delete('City')
D. df.drop(index='City')
✓ *Answer: B*

Q29. Matplotlib was created in 2003 by:

A. Peter Wang
B. Guido van Rossum
C. John D. Hunter

D. Travis Oliphant
✓ *Answer: C*

Q30. Which Matplotlib function is used to create a scatter plot?

A. plt.plot()
B. plt.bar()
C. plt.scatter()
D. plt.hist()
✓ *Answer: C*

Q31. In a Box Plot, what does the line inside the box represent?

A. Minimum (Q1)
B. Maximum (Q3)
C. Mean
D. Median (Q2)
✓ *Answer: D*

Q32. Which parameter ensures a pie chart is drawn as a perfect circle?

A. plt.grid(True)
B. plt.legend()
C. plt.axis('equal')
D. plt.show()
✓ *Answer: C*

Q33. SciPy is primarily built on top of which other library?

A. Pandas
B. NumPy
C. Matplotlib
D. OpenCV
✓ *Answer: B*

Q34. Which SciPy module contains statistical distributions and functions?

A. scipy.integrate
B. scipy.optimize
C. scipy.linalg
D. scipy.stats
✓ *Answer: D*

Q35. What does the cv2.IMREAD_GRAYSCALE flag do in OpenCV?

A. Loads a color image
B. Loads an image in grayscale mode
C. Includes the alpha channel
D. Resizes the image
✓ *Answer: B*

Q36. Which OpenCV function is used to wait for a key press?

A. cv2.imshow()
B. cv2.waitKey()
C. cv2.imwrite()
D. cv2.destroyAllWindows()
✓ *Answer: B*

Q37. What does cv2.imwrite(filename, image_array) do?

A. Displays an image
B. Destroys an image window
C. Saves an image to a file
D. Resizes an image
✓ *Answer: C*

Q38. When slicing an image array like img[0:200, 0:200] in OpenCV, what operation is being performed?

A. Resizing
B. Shading
C. Bounding Box
D. Cropping
✓ *Answer: D*

Q39. Which OpenCV function is used to draw a rectangular boundary around an object?

A. cv2.putText()
B. cv2.rectangle()
C. cv2.resize()
D. cv2.crop()
✓ *Answer: B*

Q40. Which Python library provides toolkits like Tkinter and PyQT?

A. GUI Programming
B. Game Development
C. Data Science
D. Network Programming
✓ *Answer: A*

Q41. What is the output of 3 + '34' in Python?

A. 37
B. 334
C. Error
D. '334'
✓ *Answer: C*

Q42. Which operator is used for Floor Division?

A. /
B. %
C. //
D. **
✓ *Answer: C*

Q43. The range(start, stop, step) function generates a sequence. If step is omitted, what is its default value?

A. 0
B. 1
C. 2
D. -1
✓ *Answer: B*

Q44. Which of the following is an immutable sequence data type?

A. List
B. Dictionary
C. Tuple
D. Set
✓ *Answer: C*

Q45. Which symbol is used to create a single-line comment in Python?

A. //
B. /*
C. #
D. --
✓ *Answer: C*

Q46. Which Augmented Assignment Operator multiplies a variable and assigns the new value to it?

A. +=
B. *=
C. /=
D. %=
✓ *Answer: B*

Q47. What will len([10, 20, 30]) theoretically represent based on Python basics?

A. 1
B. 2
C. 3
D. 4
✓ *Answer: C*

Q48. If df.empty returns True, what does it mean?

A. The DataFrame has missing values
B. The DataFrame has no elements
C. The DataFrame has one column
D. The DataFrame is full
✓ *Answer: B*

Q49. In Matplotlib, which function adds a descriptive label to the horizontal axis?

A. plt.title()
B. plt.ylabel()
C. plt.legend()
D. plt.xlabel()
✓ *Answer: D*

Q50. Which SciPy module is used for Fast Fourier Transform routines?

A. scipy.fftpack
B. scipy.integrate
C. scipy.interpolate
D. scipy.optimize
✓ *Answer: A*

SECTION B: Short Answer & Conceptual Questions

Q51. Define a Virtual Environment in Anaconda.

Answer: A tool to create a unique, isolated environment for a project, allowing different projects to have different dependencies and Python versions without conflicts.

Q52. What is the purpose of the 'Kernel' in a Jupyter Notebook?

Answer: It is the computational engine that executes the code in cells, maintaining the state of variables and computations across cells.

Q53. Why is proper indentation crucial in Python?

Answer: Python uses indentation to define code blocks (loops, conditionals, functions) instead of using braces. Improper indentation causes syntax or logical errors.

Q54. Differentiate between implicit and explicit type conversion.

Answer: Implicit: Automatically performed by Python (e.g., int + float = float). Explicit: Manually performed by the programmer using functions like int(), float(), str().

Q55. List any four Python Keywords.

Answer: if, else, while, True, False, for, in, return, def, class, import, from (any four)

Q56. Explain the function of the % (Modulo) operator.

Answer: Returns the remainder of a division operation. For example, 10 % 3 = 1.

Q57. Give one example of a Python Dictionary structure.

Answer: person = {'name': 'Bob', 'age': 30, 'city': 'Paris'}

Q58. What is the purpose of the end parameter in the print() function?

Answer: Specifies the character or string appended at the end of the output (default is newline '
').

Q59. Why is it discouraged to write multiple statements on a single line separated by semicolons?

Answer: It reduces code readability and makes debugging more difficult.

Q60. Name two libraries used for GUI Programming in Python.

Answer: Tkinter, PyQT, wxPython, Kivy (any two)

Q61. State the primary difference between a Module and a Package.

Answer: A Module is a single Python file (.py) containing code. A Package is a directory containing multiple modules and an __init__.py file.

Q62. What does the NLTK package stand for and what is it used for?

Answer: Natural Language Toolkit; used for working with human language data, text processing, and NLP tasks.

Q63. Define a Pandas Series.

Answer: A one-dimensional labeled array capable of holding any data type, similar to a column in a spreadsheet.

Q64. How is a DataFrame different from a Series in Pandas?

Answer: A Series is 1-dimensional with a single column of data. A DataFrame is 2-dimensional with rows and columns (tabular data).

Q65. Write the command used to install Pandas using pip.

Answer: pip install pandas

Q66. What is a Histogram?

Answer: A graphical representation of numerical data distribution showing frequency of data points in bins or intervals.

Q67. State the formula for calculating the Mean as used in SciPy.

Answer: Mean = Sum of all values / Number of values

Q68. What is OpenCV primarily aimed at?

Answer: Real-time computer vision and image processing applications.

Q69. Mention the three flags used with the imread() function in OpenCV.

Answer: IMREAD_COLOR (1) - Loads color image, IMREAD_GRAYSCALE (0) - Loads image in grayscale, IMREAD_UNCHANGED (-1) - Loads image including alpha channel

Q70. What does the cv2.putText() function do?

Answer: Adds text annotation to an image at a specified position with specified font, color, and size.

SECTION C: Application, Snippets & Code Analysis

Q71. Write the Anaconda Prompt command to create a virtual environment named ai_project using Python 3.7.2.

Answer: conda create -n ai_project python=3.7.2

Q72. Predict the output: print(12 - 5 * 6 + 2)

Answer: -16 (12 - 30 + 2 = -16)

Q73. Predict the output: print(2024 % 5)

Answer: 4

Q74. Write a Python snippet to take a user's age as input and convert it into an integer.

Answer: age = int(input('Enter your age: '))

Q75. Predict the output of: a = 10; b = 5.2; c = 38; print(a + b + c)

Answer: 53.2

Q76. Write a for loop using the range() function to print numbers from 1 to 5.

```
for i in range(1, 6): print(i)
```

Q77. Predict the output of: print([1, 2] * 3)

Answer: [1, 2, 1, 2, 1, 2]

Q78. Convert the float 5.9 to an integer explicitly and write the output.

Answer: int(5.9) -> 5

Q79. If x = 10 and y = 11, what will the expression x > y return?

Answer: False

Q80. Write the code to create a multi-line comment in Python.

```
''' This is a multi-line comment '''
```

Q81. Write a NumPy snippet to create an array with numbers ranging from 10 to 30 with a step of 5.

Answer: np.arange(10, 30, 5)

Q82. If arr = np.array([[1, 2, 3], [4, 5, 6]]), what will arr.ndim return?

Answer: 2 (It is a 2-dimensional array)

Q83. Using NumPy, how would you find the index of the minimum element in an array named arr?

Answer: np.argmin(arr)

Q84. Write a Pandas snippet to create a Series from a list [85, 92, 78] with a custom index ['Math', 'Science', 'English'].

Answer: pd.Series([85, 92, 78], index=['Math', 'Science', 'English'])

Q85. In a DataFrame df, write the command to add a new row at the end with data ['David', 40, 'Berlin'].

Answer: df.loc[len(df)] = ['David', 40, 'Berlin']

Q86. Write the code to modify a specific row in a DataFrame df to change the 'Age' at index 1 to 31.

Answer: df.loc[1, 'Age'] = 31

Q87. Write the Matplotlib code to plot a basic line chart given x = [1,2,3] and y = [2,4,6], and display it.

```
plt.plot([1,2,3], [2,4,6]) plt.show()
```

Q88. Find the median of this dataset: [45, 67, 89, 34, 56, 41, 79].

```
Sorted: 34, 41, 45, 56, 67, 79, 89 Median: 56
```

Q89. Write the OpenCV code to load an image named photo.jpg in grayscale mode.

Answer: img = cv2.imread('photo.jpg', 0)

Q90. Write the OpenCV code to apply a black shade to a specific portion of img_color from coordinates 100 to 200 on both axes.

Answer: img_color[100:200, 100:200] = (0, 0, 0)

SECTION D: Long Answer & Multi-Part Questions

Q91. A data scientist needs to work on two different AI projects simultaneously. Project A requires Python 3.6, and Project B requires Python 3.7. Explain how virtual environments solve this problem. Outline the exact steps and commands used in the Anaconda Prompt to create and activate a virtual environment for Project B.

Virtual environments create isolated spaces for each project with their own Python version and package dependencies. This prevents version conflicts.
Commands:
conda create -n project_b python=3.7
conda activate project_b

Q92. Analyze the Python statement: if a > b: Explain the role of indentation. What happens if the developer forgets to indent? Contrast Python's code blocks with C++/Java.

> *Answer:* Indentation tells Python which statements belong to the if block. Without indentation, Python raises IndentationError. Python uses indentation while C++/Java use curly braces {}.

Q93. A student writes: count = 0; while count < 5: print(count). Identify the logical flaw, state what this is called, and write the corrected version.

The loop lacks an update statement (count += 1). This is called an Infinite Loop.
Corrected:

```
count = 0
while count < 5:
    print(count)

    count += 1
```

Q94. Compare NumPy arrays with standard Python lists based on Homogeneity, Memory Efficiency, Performance, and Functionality.

Homogeneity: NumPy is homogeneous, lists are heterogeneous
Memory: NumPy is more memory efficient
Performance: NumPy is faster for numerical operations
Functionality: NumPy has built-in math functions, lists don't

Q95. A NumPy array is created: a = np.array([10, 20, 30]). Write the code and output if you subtract 2 from every element, and if you perform exponentiation raising each element to the power of 2.

a - 2 -> [8, 18, 28]
a ** 2 -> [100, 400, 900]

Q96. A teacher wants to create a digital record of three students using a Pandas DataFrame. Include Name, Age, and City. Write the code to create a dictionary, convert to DataFrame, and delete the 'City' column.

```
data = {'Name': ['Alice', 'Bob', 'Charlie'], 'Age': [25, 30, 35], 'City': ['New York', 'Paris', 'London']}
df = pd.DataFrame(data)
df.drop(columns='City', inplace=True)
```

Q97. Explain the fundamental difference between a Pandas Series and a Pandas DataFrame. Provide an example of how positional indexing differs from label indexing.

Series is 1-dimensional (single column), DataFrame is 2-dimensional (table).
Positional: s[0] (first element)
Label: s['Math'] (element with label 'Math')

Q98. A marketing team wants to visualize their monthly sales. Identify the Matplotlib function used to create a vertical Bar Chart. Write the Python snippet to create a bar chart for categories ['Jan', 'Feb', 'Mar'] with values [15, 24, 10].

Function: plt.bar()
Code:

```
plt.bar(['Jan', 'Feb', 'Mar'], [15, 24, 10])
plt.xlabel('Months')
plt.ylabel('Sales')
plt.title('Monthly Sales')
plt.show()
```

Q99. A developer is building an AI vision system to crop faces from photos. Write OpenCV code to read an image, crop a ROI from (0,0) to (200,200), and display the cropped image.

```
img = cv2.imread('face.jpg', 1)
roi = img[0:200, 0:200]
cv2.imshow('Cropped', roi)
cv2.waitKey(0)
```

Q100. Explain the purpose of the cv2.rectangle() function in OpenCV. Detail what parameters must be passed.

Draws a rectangular boundary around an object.
Parameters: image, top-left (x1,y1), bottom-right (x2,y2), color (B,G,R), thickness

QUICK REVISION NOTES

Topic	Key Points
Environments & Notebooks	Anaconda (2012, Wang & Oliphant), Virtual Env, Jupyter (localhost:8888)
Python Basics	Indentation, Comments (#, """), Identifiers, Operators (//, **, %), Data Types
NumPy	Multi-dimensional arrays, Fast, Memory-efficient, np.array(), arange(), mean(), argmax
Pandas	Series (1-D), DataFrame (2-D), df.loc[], df.drop()
Matplotlib	Line plot, bar, scatter, hist, pie, boxplot
SciPy	Built on NumPy, stats, integrate, optimize
OpenCV	Real-time vision, imread(), imshow(), imwrite(), resize()